

SEPARATIONS

Separation of Ethyl Acetate–Ethanol Azeotropic Mixture Using Hydrophilic Ionic Liquids

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The separation of ethyl acetate and ethanol (EtOH) is important but difficult due to their close boiling points and formation of an azeotropic mixture. The separation of the azeotropic mixture of ethyl acetate and EtOH using the hydrophilic ionic liquids (ILs) 1-alkyl-3-methylimidazolium chloride (alkyl = butyl, hexyl, and octyl) ([C_nmim]Cl, *n* = 4, 6, 8) and 1-allyl-3-methylimidazolium chloride and bromide ([Amim]Cl and [Amim]Br) has been investigated. Triangle phase diagrams of five ILs with ethyl acetate and EtOH were constructed, and the biphasic regions were found as follows: [Amim]Cl > [Amim]Br > [C₄mim]Cl > [C₆mim]Cl > [C₈mim]Cl. The mechanisms of the ILs including cation, anion, and polarity effect were discussed. The results showed that the hydrophilic ILs [C_nmim]Cl (*n* = 4, 6, 8), [Amim]Br, and [Amim]Cl could remove EtOH effectively from the azeotropic mixture of ethyl acetate and EtOH. Moreover, it was found that [Amim]Cl had the highest extraction efficiency, and the purity of ethyl acetate could reach 99.27 wt % after extraction twice. These hydrophilic ILs are easily synthesized and purified, are economically feasible, and caused no erosion to the equipment, which usually happened for ILs containing F. ILs could be recycled by simple distillation. The separating process can reduce the energy consumption greatly, and the total process is green and environmentally benign.

1. Introduction

Ionic liquids (ILs) have been widely recognized as green and potential environmentally friendly solvents because of their negligible vapor pressure, nonflammability, thermal stabilization, high dissolving ability for both polar and nonpolar compounds,^{1–3} and many variations of both cation and anion conformations.⁴ ILs have been studied for chemical synthesis,^{5–7} controlled processing of polymer materials,⁸ versatile electrolytes for diverse technologies,^{9–14} catalysis, and extraction processes.^{15–26}

The imidazolium-based ILs are highly ordered hydrogen-bonded solvents and have strong solvent effects on chemical reactions and processes. Thus, the ability of a solute to form hydrogen bonds or other possible interactions with potential solvents is an important feature of their behavior.²⁷ Therefore, using imidazolium-based ILs as replacements for the traditional volatile organic compounds (VOCs) to separate azeotropic mixtures and as an alternative to conventional extractive distillation has been investigated widely; examples of such ILs are shown in Table 1.

Although imidazolium-based ILs have shown a wide range of application in azeotropic mixture separation, some practical drawbacks cannot be neglected. First, the synthesis processes of some ILs listed in Table 1 are complex, such as [C₂mim]-[EtSO₄]. Both the hydrophilic and hydrophobic ILs listed in Table 1 are prepared following standard procedures by metath-

Table 1. Examples of ILs Used in the Liquid–Liquid Separation and Extraction of Mixture Systems

mixture systems	types of IL	ref
toluene and heptane	[C ₂ mim]I ₃ , [C ₄ mim]I ₃	19
gasoline octane boosters	[C ₂ mim][EtSO ₄]	21
ethanol and heptane	[C ₆ mim][PF ₆]	23
1-hexene and <i>trans</i> -3-hexene	[C ₄ mim][PF ₆], [iBemim][BF ₄]	25
aromatic hydrocarbons and alkanes	[C ₂ mim][NTf ₂]	26

esis or ion exchange of ILs containing halide anion with corresponding anions, which result in higher preparation prices.⁴ Second, ILs are required to be biodegradable or incinerable. In both cases F materials are undesirable.²⁸ Finally, the higher viscosities of the hydrophobic ILs make the operating processes discommodious in practice. Compared with the ILs listed in Table 1, the hydrophilic ILs containing chloride and bromide anions, such as 1-alkyl-3-methylimidazolium chloride (alkyl = butyl, hexyl, and octyl) ([C_nmim]Cl; *n* = 4, 6, 8) and 1-allyl-3-methylimidazolium chloride and bromide ([Amim]Cl and [Amim]Br) in this investigation, are easily prepared through one simple step, and are relatively lower in price and environmentally acceptable. The excellent properties of the hydrophilic ILs make them have extensive and practical applications in cellulose dissolution^{29,30} and azeotropic mixture separation.^{31,32}

The boiling points of ethyl acetate, EtOH, and their mixture are 78.5, 77.06, and 71.8 °C, respectively. EtOH cannot be separated from the mixture of ethyl acetate–EtOH by simple distillation or rectification due to their close boiling points and formation of an azeotropic mixture. The purification techniques of ethyl acetate have been investigated, such as azeotropic distillation,³³ extractive distillation,³⁴ extractive dehydration,³⁵ and membrane separation.³⁶ Although the techniques have been

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used for some practical progress, some disadvantages should be noted. The separation processes are relatively complicated, and the environmental pollution from adding new VOCs should not be neglected in azeotropic distillation and extraction distillation. High-energy consumption for recycling the salt and serious corrosion to the equipment because of added salt cannot be avoided in extractive dehydration. The regeneration of the membrane is difficult. All of these are handicaps to the purification of ethyl acetate in industrial production.

Mutelet et al. and Heintz et al. have reported thermodynamic properties of solutes in ILs and activity coefficients at infinite dilution of polar solutes in ILs using inverse gas chromatography, gas-liquid chromatography, and the transpiration method, respectively.^{37–49} Their investigations provided systematic data, which were believed to be very important for selecting the appropriate ILs in different applications of chemical engineering and significant for understanding the behavior of ILs and the interactions between solutes and ILs. However, the investigations were limited to large numbers of binary systems including ILs, but few ternary systems, especially such as the system of IL-ethyl acetate-EtOH.

In this paper, we investigated the separation of ethyl acetate and EtOH using the hydrophilic ILs 1-alkyl-3-methylimidazolium chloride (alkyl = butyl, hexyl, and octyl) ([C_nmim]Cl; *n* = 4, 6, 8) and 1-allyl-3-methylimidazolium chloride and bromide ([Amim]Cl and [Amim]Br) in IL-ethyl acetate-EtOH ternary mixtures and discussed the mechanics of the separation. The hydrophilic ILs have been proposed as promising solvents for separating the azeotropic mixture of ethyl acetate and EtOH, and it is possible to obtain highly pure ethyl acetate without much energy consumption. Ethyl acetate and EtOH in IL phase were easily removed by distillation, and IL could be recycled for extractive process again.

2. Materials and Methods

[C₄mim]Cl, [C₆mim]Cl, [C₈mim]Cl, [Amim]Br, and [Amim]Cl were synthesized, purified, and dried in a vacuum according to previously published methods^{4,30} and characterized by ¹H/¹³C NMR.

[C₄mim]Cl: ¹H NMR (400 MHz; DMSO; 25 °C): δ (ppm) = 0.88 (3H, t), 1.24 (2H, m), 1.75 (2H, m), 3.87 (3H, s), 4.18 (2H, t, *J* = 3.2 Hz), 7.77 (1H, s), 7.85 (1H, s), 9.45 (1H, s). ¹³C NMR (100 MHz; DMSO, 25 °C): δ (ppm) = 13.21, 18.69, 31.33, 35.63, 48.31, 122.21, 123.49, 136.71.

[C₆mim]Cl: ¹H NMR (600 MHz; DMSO; 25 °C): δ (ppm) = 0.86 (3H, m), 1.29 (6H, m), 1.87 (2H, m), 4.07 (3H, s), 4.29 (2H, t, *J* = 7.2 Hz), 7.46 (1H, s), 7.65 (1H, s), 9.95 (1H, s). ¹³C NMR (125 MHz; DMSO, 25 °C): δ (ppm) = 13.76, 22.20, 25.72, 30.02, 30.94, 36.41, 49.78, 121.71, 121.73, 123.69, 137.15.

[C₈mim]Cl: ¹H NMR (400 MHz; DMSO; 25 °C): δ (ppm) = 0.768–0.802 (3H, t, *J* = 7.2 Hz), 1.187–1.234 (12H, m), 1.779–1.813 (2H, t, *J* = 6.8 Hz), 3.819 (3H, s), 4.099–4.134 (2H, t, *J* = 7.2 Hz), 7.356 (1H, s), 7.39 (1H, d, *J* = 1.6 Hz), 8.63 (1H, s). ¹³C NMR (100 MHz; DMSO, 25 °C): δ (ppm) = 13.84, 22.12, 25.54, 28.40, 28.53, 29.43, 31.24, 35.65, 48.88, 122.23, 123.56, 136.53.

[Amim]Br: ¹H NMR (400 MHz; DMSO; 25 °C): δ (ppm) = 3.868 (3H, s, N-CH₃), 4.846–4.861 (2H, d, *J* = 6.0 Hz, N-CH₂-CH=CH₂), 5.275–5.373 (2H, m, N-CH₂-CH=CH₂), 5.991–6.074 (1H, m, N-CH₂-CH=CH₂), 7.723–7.742 (2H, s, N-CH=CH-N), 9.160 (1H, s, N-CH-N). ¹³C NMR (100 MHz; CDCl₃, 25 °C): δ (ppm) = 35.86 (N-CH₃), 50.57 (N-CH₂-CH=CH₂), 120.15 (N-CH₂-CH=CH₂), 122.17 (N-

CH₂-CH=CH₂), 123.61 (N-CH-CH-N), 131.72 (N-CH-CH-N), 136.49 (N-CH-N).

[Amim]Cl: ¹H NMR (400 MHz; DMSO; 25 °C): δ (ppm) = 3.883 (3H, s, N-CH₃), 4.877–4.888 (2H, d, *J* = 4.4 Hz, N-CH₂-CH=CH₂), 5.277–5.354 (2H, m, N-CH₂-CH=CH₂), 5.992–6.075 (1H, m, N-CH₂-CH=CH₂), 7.771–7.787 (2H, s, N-CH-CH-N), 9.374 (1H, s, N-CH-N). ¹³C NMR (100 MHz; CDCl₃, 25 °C): δ (ppm) = 35.71 (N-CH₃), 50.53 (N-CH₂-CH=CH₂), 120.00 (N-CH₂-CH=CH₂), 122.22 (N-CH₂-CH=CH₂), 123.66 (N-CH-CH-N), 131.87 (N-CH-CH-N), 136.76 (N-CH-N).

Ethyl acetate and EtOH were all of analytical grade and at least 99% pure (Beijing Chemical, Beijing, China). IL-ethyl acetate-EtOH ternary diagrams were determined by the cloud-point method. Ethyl acetate was added dropwise to the known amount of IL in the vessel until the mixture became cloudy. EtOH was added dropwise to the vessel again to get a clearly monophasic system. The composition of this mixture was recorded. The additional points were added repeatedly and the monophasic system formed at last. The weight percentages of EtOH and ethyl acetate in the IL phase were determined by HPLC, which was equipped with a pump (515 HPLC Pump, Waters), a refractive index (RI) detector (Waters 2414), and a SunFire C₁₈ analytical column (250 mm length $\times$ 4.6 mm i.d.; 5 μ m particle size). The RI detector temperature was kept at 30 °C. All experiments were conducted at 25 °C.

To determine the extraction efficiency of EtOH and weight percent ethyl acetate in the IL phase, samples in the separating funnels containing 2 mL of IL and an equal volume mixture of ethyl acetate and EtOH with known composition were used for analysis.⁵⁰ The mixtures were vibrated for 3 min, and then held still for 30 min to equilibrate the formed two phases. The two phases were separated respectively and diluted to 100 mL using the mobile phase of HPLC, composed of chromatographic methanol and pure water in a certain proportion. All the peaks of IL, ethyl acetate, and EtOH were separated without any overlap. ILs can flow out with the mobile phase and will not harm the chromatographic column. The relative composition ratios of ethyl acetate and EtOH were determined by measuring the relative peak areas of the detector signals.

3. Results and Discussion

3.1. Triangular Phase Diagrams. Ethyl acetate was found to be nearly immiscible with [C₄mim]Cl, [Amim]Cl, and [Amim]Br, and partially miscible with [C₆mim]Cl and [C₈mim]Cl at 25 °C. The solubility was found as follows: [C₄mim]Cl ([Amim]Cl, [Amim]Br) < [C₆mim]Cl < [C₈mim]Cl. At 25 °C, all five hydrophilic ILs were found to be totally miscible with EtOH at all proportions.

Ternary systems of ethyl acetate and EtOH with all five hydrophilic ILs were investigated. The ternary phase diagrams are shown in Figure 1a as the weight ratios of the three components and Figure 1b as the mole fractions of the three components. Ethyl acetate and [C₄mim]Cl, [Amim]Br, and [Amim]Cl were mutually nearly immiscible, and the binodal curve was increased from 0 to 100 along the [C₄mim]Cl ([Amim]Br and [Amim]Cl)-ethyl acetate axis in the weight percent phase diagram. [C₄mim]Cl is solid at room temperature and did not dissolve until the mole fraction of EtOH up to 0.0816 in the [C₄mim]Cl-ethyl acetate-EtOH system. The solubility of ethyl acetate in [C₆mim]Cl and [C₈mim]Cl was up to 27.94 and 35.95 wt % in the ternary system, respectively, corresponding to the mole fractions of 0.4718 and 0.5954. With increasing amount of ethyl acetate, a biphasic system of IL-rich bottom

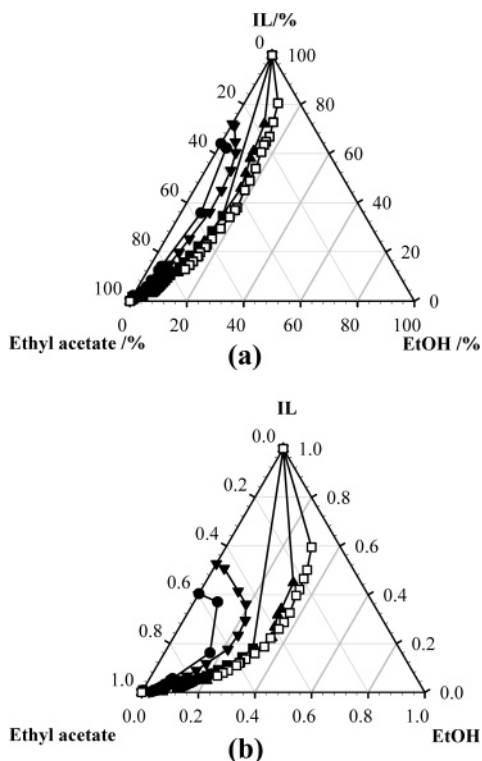


Figure 1. Triangular phase diagrams for ternary compositions of [Amim]-Cl, [Amim]Br, [C₄mim]Cl, [C₆mim]Cl, and [C₈mim]Cl with ethyl acetate and EtOH plotted as weight ratios (a) and mole fractions (b), respectively, determined at 25 °C. [C₄mim]Cl, —■—; [C₆mim]Cl, —▼—; [C₈mim]Cl, —●—; [Amim]Br, —▲—; and [Amim]Cl, —□—. The regions to the left of (—■—, —▼—, —●—, —▲—, —□—) are biphasic with ethyl acetate rich and IL-rich phases. The regions extending from the ethyl acetate and EtOH axis to IL apex are the monophasic composition containing all three components.

phase and ethyl acetate rich top phase was formed. The biphasic region extended along the ethyl acetate–IL axis, and the critical point moved to higher weight percent EtOH. The biphasic regions were found as follows: [Amim]Cl > [Amim]Br > [C₄mim]Cl > [C₆mim]Cl > [C₈mim]Cl. The presence of EtOH increased the miscibility of ethyl acetate with ILs, and monophasic system could be obtained when the mole fraction of EtOH in ethyl acetate and EtOH mixtures was up to 0.4011, 0.2752, 0.0900, 0.3886, and 0.5531 in the system of [C₄mim]Cl, [C₆mim]Cl, [C₈mim]Cl, [Amim]Br, and [Amim]Cl, respectively. The completely miscible region of the three components (IL, ethyl acetate, and EtOH) was shown between the binodal curves and extended along the ethyl acetate and EtOH axis.

The effect of the cations and the anions of the ILs should be considered for the formation of biphasic regions. The cation was important for the system. van der Waals interactions existed between the alkyl chain of the imidazolium ring and the alkyl chain of EtOH, and hydrogen bonding existed between the C₂-H, C₄-H, C₅-H, and EtOH, forming an H–O···H chain. Moreover, the hydrogen bonding between the C₂-H and EtOH was stronger than those of the others, and there were much weaker interactions between ethyl acetate and IL than between EtOH and IL. Thus EtOH was much more favorable to distribute to the IL phase. With increasing length of the alkyl chain of the imidazolium ring, IL formed larger neutral aggregates;⁵¹ therefore, the interactions including anion–EtOH and the hydrogens of the alkyl chain–EtOH became weak. The possible reason was that anion integrated with cation by Coulombic interaction, and the steric effect increased as the length of the alkyl chain increased. Consequently, the integration of anion

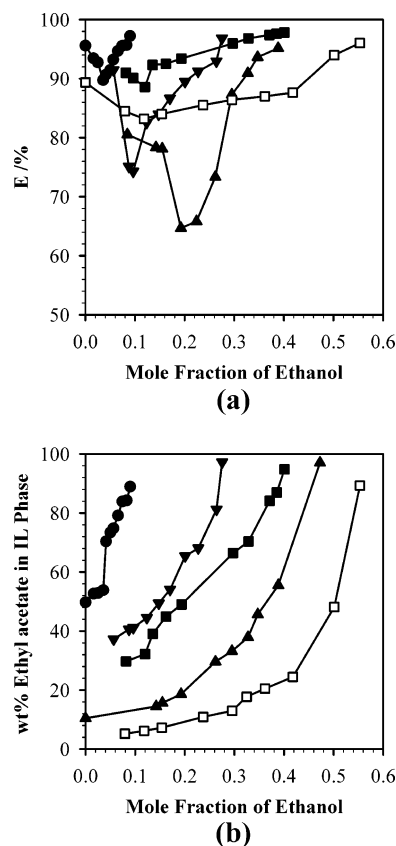
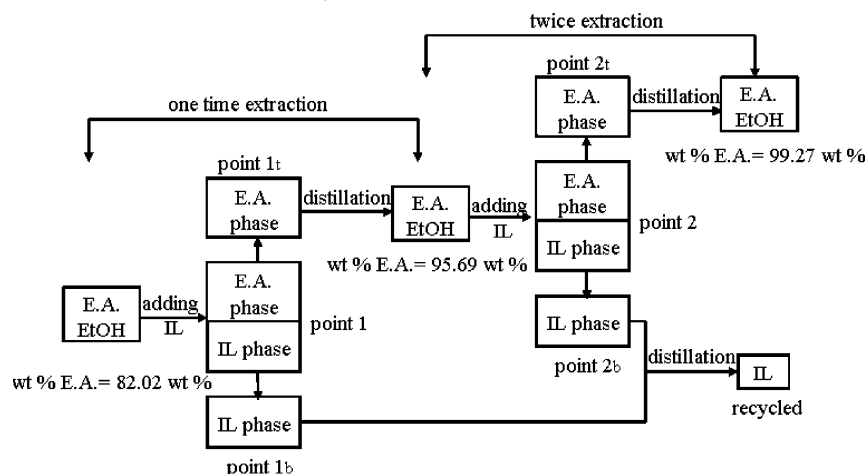


Figure 2. Extraction efficiency of EtOH in IL–ethyl acetate–EtOH systems for [C₄mim]Cl, [C₆mim]Cl, [C₈mim]Cl, [Amim]Br, and [Amim]Cl as a function of initial mole fraction of EtOH in the ethyl acetate phase (a), measured at 25 °C. [C₄mim]Cl, —■—; [C₆mim]Cl, —▼—; [C₈mim]Cl, —●—; [Amim]Br, —▲—; and [Amim]Cl, —□—. With increasing mole fraction of EtOH, weight percent ethyl acetate (b) increased all the time until the monophasic region formed. The diagrams indicate the abilities of ILs to extract EtOH and the miscibility of ethyl acetate in ILs, respectively.

with EtOH was difficult, and the biphasic regions were as follows: [C₄mim]Cl > [C₆mim]Cl > [C₈mim]Cl. Compared with [C_{*n*}mim]⁺, [Amim]⁺ had a smaller ion size due to three carbon atoms and a double bond in the N-substituted methimidazolium cation of [Amim]⁺.³⁰ It was suggested that the steric effect was smaller than that for [C_{*n*}mim]⁺, and the anion could interact with EtOH actively. On the other hand, the π electrons of the double bond in the allyl had interactions with the imidazolium ring; thus the total cation and anion were more active and the ability of integration with ethyl acetate and EtOH also increased. As a result, the biphasic regions of [Amim]Cl and [Amim]Br were relatively greater than that of [C_{*n*}mim]Cl (*n* = 4, 6, 8).

The anion played a key role in the systems. The imidazolium-based ILs and EtOH had strong hydrogen bonding between the anion, for example, Cl[−], and EtOH and formed an O–H···Cl chain. On the other hand, the hydrogen bonding also existed between Cl[−] and the hydrogens of the alkyl chain, with the C₂-H, C₄-H, and C₅-H of the imidazolium ring forming an H···Cl chain, too. Moreover, the hydrogen bonding between the C₂-H and Cl[−] was stronger than others. The reasonable explanation was that the presence of the two N atoms and the positive charge in the imidazolium ring made an acidic C₂-H bond between them;⁵² consequently, the [C₂-H···O–H] interaction possessed more negative interaction energy. With increasing mole fraction of EtOH, the interaction between EtOH and the hydrogens of the imidazolium ring became stronger. The interaction of Br[−] with EtOH and the hydrogens of the

Scheme 1. Flowchart of Extraction Processes of Ethyl Acetate^a

^a E.A., ethyl acetate; IL, [Amim]Cl; points 1 and 2, mixtures of IL, ethyl acetate, and EtOH; 1_t, 1_b, 2_t, and 2_b, top and bottom phases of points 1 and 2, respectively.

imidazolium ring became weaker than that of Cl⁻. The possible reason was that the radius of Cl⁻ was smaller than that of Br⁻; therefore, the solvating action of Cl⁻ was stronger than that of Br⁻. In other words, the interaction between EtOH and Cl⁻ was stronger than that between EtOH and Br⁻. Therefore, the biphasic region of [Amim]Cl was greater than that of [Amim]Br. As a result, the biphasic regions of the five systems were as follows: [Amim]Cl > [Amim]Br > [C₄mim]Cl > [C₆mim]Cl > [C₈mim]Cl.

3.2. Extraction Efficiency. In each of the ternary systems, with increasing mole fraction of EtOH (χ_{EtOH}) in the initial mixture of ethyl acetate and EtOH, the extraction efficiency of EtOH (Figure 2a) decreased first and then increased, and the weight percent ethyl acetate in the IL phase (Figure 2b) increased in the biphasic regions from $\chi_{\text{EtOH}} = 0$ to $\chi_{\text{EtOH}} = 0.60$.

The extraction efficiency, E %, indicates the ability of [C₄mim]Cl, [C₆mim]Cl, [C₈mim]Cl, [Amim]Br, and [Amim]Cl to remove EtOH from the azeotropic mixture of ethyl acetate and EtOH, and is defined as follows:

$$E \% = \frac{C_{\text{IL phase}} V_{\text{IL phase}}}{C_{\text{mix}} V_{\text{mix}}} \times 100$$

where $C_{\text{IL phase}}$ and $V_{\text{IL phase}}$ are the concentration and volume of EtOH in the IL phase, respectively. C_{mix} and V_{mix} are the concentration and volume of EtOH in the initial mixture of ethyl acetate and EtOH, respectively.

The biphasic region was greatest for [Amim]Cl, then [Amim]Br, and decreased with increasing the alkyl length of the imidazolium ring from C₄ to C₈. A similar rule was found by Rogers et al.,⁵⁰ who have reported the solubilization of water with ethanol in [C_{*n*}mim][PF₆] ($n = 4, 6, 8$). In the systems of [C_{*n*}mim][PF₆] ($n = 4, 6, 8$)–H₂O–EtOH, the biphasic region was increased from C₄ to C₈, but the distribution of EtOH and H₂O content in the IL phase was greatest for [C₄mim][PF₆] and decreased with increasing the alkyl length of the imidazolium ring from C₄ to C₈. In our investigation, it was found that EtOH preferred the IL phase and IL could remove most EtOH from the mixtures of EtOH and ethyl acetate, leaving a high-purity sample of ethyl acetate. The extraction efficiency of EtOH in IL phase decreased first and then increased with increasing mole fraction of EtOH up to 0.1199, 0.0964, 0.0356, 0.1927, and 0.1181 in the system of [C₄mim]Cl, [C₆mim]Cl, [C₈mim]Cl,

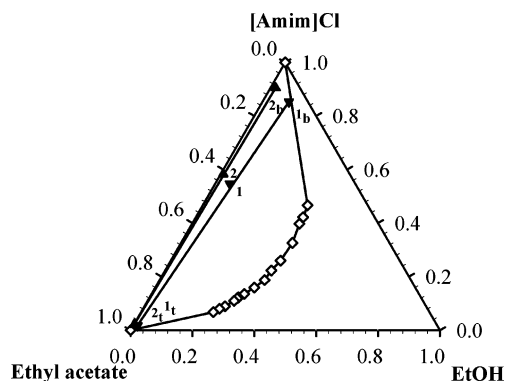


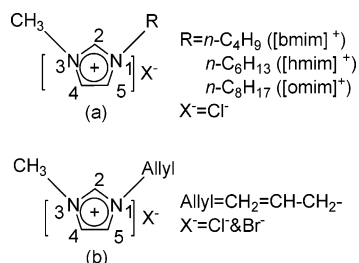
Figure 3. Experimental tie lines (—▼— and —▲—) for twice extraction processes of ethyl acetate of the ternary system [Amim]Cl + ethyl acetate + EtOH at 25 °C, shown in the ternary phase diagram of [Amim]Cl–ethyl acetate–EtOH (—□—). Points 1 and 2: mixture of IL, ethyl acetate, and EtOH. 1_t, 1_b, 2_t, and 2_b: top and bottom phases of points 1 and 2, respectively.

[Amim]Br, and [Amim]Cl, respectively. With increasing mole fraction of EtOH, the weight percent ethyl acetate in the IL phase increased to 94.80, 97.22, 88.93, 97.03, and 89.28 wt % for the five IL–ethyl acetate–EtOH systems and formed a monophasic system at last.

In the extraction process, IL is added as the extractor of the mixture. The separation makes most high-purity ethyl acetate in the ethyl acetate phase, leaving IL and EtOH with a small amount of ethyl acetate in the IL phase. The higher purity ethyl acetate could be separated repeatedly using this method. IL could be recycled by distillation. For example, the flowchart of extraction processes of ethyl acetate in the system of [Amim]Cl–ethyl acetate–EtOH is shown in Scheme 1. A sample containing 82.02 wt % ethyl acetate in the initial mixture of ethyl acetate and EtOH was selected randomly for the demonstration. According to the mole ratio measured of EtOH and ethyl acetate in the top phase and bottom phase, it was possible to calculate that ethyl acetate weight percent became 95.69 wt % after extraction once by adding an equal volume of [Amim]Cl to the mixture of ethyl acetate and EtOH, and it could be up to 99.27 wt % after extraction twice using this method. The points of azeotropic mixtures and the corresponding top and bottom phases measured using HPLC are shown in Figure 3, corresponding to points 1, 2, 1_t (top phase of point 1), 1_b (bottom phase of point 1), 2_t (top phase of point 2), and 2_b (bottom phase

Table 2. Mole Fraction of Ethyl Acetate, EtOH, and IL Corresponding to Points 1, 2, 1_t, 1_b, 2_b, and 2_b

	point 1			point 2		
	χ_{EtOH}	χ_{IL}	$\chi_{\text{ethyl acetate}}$	χ_{EtOH}	χ_{IL}	$\chi_{\text{ethyl acetate}}$
mixture	0.0471	0.5469	0.4060	0.0082	0.5829	0.4089
top phase	0.0016	0.0212	0.9772	0.0137	0.0175	0.9688
bottom phase	0.0167	0.9046	0.0787	0.0856	0.8531	0.0613

**Figure 4.** Structural variation in the 1-alkyl-3-methylimidazolium chloride salts (a) and 1-allyl-3-methylimidazolium chloride and bromide salts (b) used in this study.

of point 2), respectively. The values of points 1, 2, 1_t, 1_b, 2_t and 2_b were shown in Table 2. However, due to the closeness to the axis of [Amim]Cl and ethyl acetate, it was also found that the small deviation of point 2 happened due to the different methods of cloud-point and HPLC determinations.

3.3. Polarity Effect. ILs composed of cation [C_nmim]⁺ or [Amim]⁺ and anion Cl⁻ or Br⁻ can act as both hydrogen-bond acceptors ([Cl]⁻ or [Br]⁻) and donors ([C_nmim]⁺ or [Amim]⁺), which would be expected to interact with solvents by both accepting and donating sites, as shown in Figure 4. In the IL–ethyl acetate–EtOH ternary system, the three components IL, ethyl acetate, and EtOH had complex connections with each other, and ILs destroyed the interaction of the azeotropic mixture of ethyl acetate and EtOH. The complex effects were explained by intermolecular interactions such as van der Waals interactions, hydrogen bonding including interspecies hydrogen bonding and intraspecies hydrogen bonding, and Coulombic interaction. However, in the systems containing ions, the polarity effect usually outweighs other effects.⁴⁶ The polarity of [C₆mim]Cl was reported to be 1.06,⁵³ and most ILs have similar polarities.⁵⁴ The polarities of ethyl acetate and EtOH are 0.656 and 0.520,⁵⁵ respectively, relatively weaker than ILs. Because of the particular character of EtOH, it could be miscible with both ethyl acetate and ILs. The more polar the solute is, the greater the interactions with the ILs are.⁴⁷ Hence, in the IL–ethyl acetate–EtOH ternary system, EtOH had greater interaction with IL than with ethyl acetate, and EtOH was extracted to the IL phase. Moreover, the presence of EtOH increased the polarity of the mixture of ethyl acetate and EtOH, and decreased the polarity of the IL because of the complex interactions among them. Hence, the mutual miscibility between ethyl acetate and ILs increased. With increasing mole fraction of EtOH, ethyl acetate and IL were miscible with each other, the biphasic system formed more slowly, and a homogeneous phase formed at last. When the mole fraction of EtOH was small, the interactions of IL and EtOH were feeble, and the interactions of ethyl acetate and EtOH were dominant. However, with increasing mole fraction of EtOH, the intraspecies hydrogen bonding between the anion and the hydrogens of the imidazolium ring became weaker. Thus the viscosity of IL decreased and the miscibility of IL and ethyl acetate increased. The biphasic system formed more slowly, and the biphasic region became smaller. The interaction between IL and EtOH became stronger. Hence, the extraction efficiency of EtOH decreased

first and then increased. On the other hand, with increasing the length of the alkyl chain of the imidazolium ring from C₄ to C₈, the miscibility of ethyl acetate and ILs increased, and the polarity and the hydrogen bond basicity of ILs decreased, which has been suggested by Mutelet et al.⁴⁷ Moreover, the parameters calculated by Mutelet et al. and the analysis from our investigation lead to the same conclusion about the interactions of the multicomponent systems. For [Amim]Br and [Amim]Cl, the allyl of the imidazolium ring made the miscibility of ethyl acetate smaller than other alkyl ILs, which suggested that the polarities of [Amim]Br and [Amim]Cl were greater than that of [C_nmim]Cl (n = 4, 6, 8). It was maybe another reason that the biphasic region became smaller and the ability of extracting EtOH from the azeotropic mixture of EtOH and ethyl acetate was as follows: [Amim]Cl > [Amim]Br > [C₄mim]Cl > [C₆mim]Cl > [C₈mim]Cl.

4. Conclusions

The hydrophilic ILs [C₄mim]Cl, [Amim]Br, and [Amim]Cl are nearly immiscible with ethyl acetate, whereas [C₆mim]Cl and [C₈mim]Cl are partially miscible with ethyl acetate. When adding EtOH to the biphasic mixtures, the miscibility of ethyl acetate increased, and a homogeneous phase formed when the concentration of IL, EtOH, and ethyl acetate reached a certain point in the triangle phase diagram. EtOH prefers the IL phase rather than the ethyl acetate phase. Of all five ILs, the biphasic region of [Amim]Cl was the largest, and the extractive ability of [Amim]Cl for EtOH was higher than those of the other ILs. [Amim]Cl was superior to the other ILs for the separation of the azeotropic mixture of ethyl acetate and EtOH. It is possible to use hydrophilic ILs to extract EtOH from the azeotropic mixture of ethyl acetate and EtOH. The results are important for the separation of the azeotropic mixture of ethyl acetate and EtOH. As extractors, the hydrophilic imidazolium-based ILs containing chloride and bromide have important applications for the azeotropic mixtures. These processes could lead to an environmentally benign extraction process for the separation of azeotropic mixtures rather than VOCs. Moreover, the ILs are economically feasible, are easily recycled by simple distillation, and do not erode the reactor. The extraction processes are green, are environmentally benign, and could be implemented in real processes.

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